

Predicting the long-term impact of AI assistants on collective intelligence

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Abstract

Recent advances in artificial intelligence (AI) have enabled widespread use of AI assistants for various cognitive tasks, ranging from simple everyday tasks to scientific discovery and creative writing. While AI assistance can improve the performance of each individual, it can weaken their subsequent performance without AI and further limit the diversity of interests and opinions. These risks raise the question of whether human collective intelligence can persist as our society becomes increasingly reliant on AI assistants. To address this question, we develop an evolutionary model of collective intelligence, where each individual participate in a prediction task, consult AI, and imitate each other's prediction strategy. We find that "Chatbot" AI, which depends on users' interests, can prevent the emergence of collective intelligence by either limiting diversity or propagating systematic bias through overreliance. Directly incentivizing or penalizing AI use generally worsens these outcomes. Instead, a balanced incentive structure that preserve minority expertise and reward contributions to collective outcome can promote the emergence of collective intelligence. However, the inherent ability of the population's to make prediction in the absence of AI remains suboptimal even under this balanced incentive structure, raising concerns for continued AI reliance. **[SWP: conclusion?]**

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability (Galton, 1907; Woolley et al., 2010; Yaniv and Milyavsky, 2007; Hommes et al., 2005; Lorge et al., 1958). This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups (Dreyer et al., 2025; Simons, 2004; Berdahl et al., 2013) and human societies (Woolley et al., 2010; Riedl et al., 2021) to machine-learning systems (Breiman, 1996, 2001). Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences (Mann and Helbing, 2017), evolutionary biology (Wang et al., 2025), and computer science (Werfel, 2026). Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes (Cui and Yasseri, 2024).

The recent emergence of generative AI and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks (Noy and Zhang, 2023; Brynjolfsson et al., 2025), including scientific discovery (Lu et al., 2026) and creative writing (Doshi and Hauser, 2024). In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing (Bick et al., 2026; Sajadieh et al., 2026), the potential long-term impact of AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise (Vaccaro et al., 2024), and opinion aggregation (Burton et al., 2024; Tessler et al., 2024). Conversely, empirical evidence indicate that the use of AI can limit the diversity of opinions (Fügener et al., 2021; Doshi and Hauser, 2024; Lee and Chung, 2024; Meincke et al., 2025), a key requirement for the emergence of CI (Hong and Page, 2004; Lorenz et al., 2011; Aplin et al., 2014; Becker et al., 2017).

Our ability to predict the long-term impact of AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI use. On one hand, many industrial sectors are pushing towards increased AI use, where the number of “tokens” used is often perceived as a proxy for productivity (Bousquette, 2026; Keary, 2026). On the other hand, many academic institutions appear to discourage, and even penalize, AI use, reflecting concerns about plagiarism, misinformation, and limited independence (Thorp, 2023; Chawla, 2026). Such discrepancy raises a question about whether directly incentivizing or penalizing AI use will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of AI assistants. Such incentive structure should seek to retain the benefits of AI and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual participates in a prediction task from partial observations of under-

69 lying causal factors (Mann and Helbing, 2017; Wang et al., 2025; Hong et al., 2012).
 70 In particular, we allow each individual to independently consult with AI assistants
 71 and revise their predictions accordingly. The resulting model allows us to investi-
 72 gate the robustness of human CI under different assumptions about the AI model’s
 73 ability to make predictions as well as incentive structures that are associated with
 74 the prediction task. In doing so, we assess the impact of directly incentivizing or
 75 penalizing AI use on CI and further suggest a balanced incentive structure that can
 76 maintain CI in the presence of AI.

77 Mathematical model

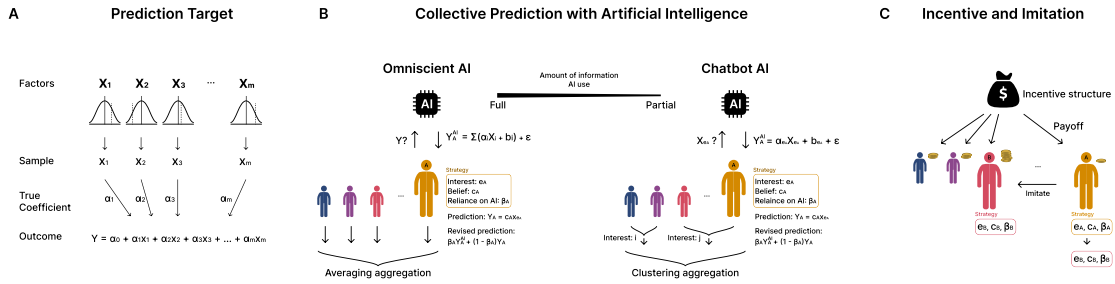


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

78 We begin by extending previously established models of CI, in which a group of
 79 players participates in a prediction task (Fig. 1): each player makes an individual
 80 prediction based on partial observations of the underlying causal factors, and the
 81 group forms a collective prediction by aggregating these individual predictions (Hong
 82 et al., 2012; Mann and Helbing, 2017; Wang et al., 2025). Specifically, the prediction
 83 target, Y , is modeled as a linear combination of M causal factors x_i , with fixed
 84 coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor is drawn from independent
 85 normal distribution (Fig. 1A). Then, each individual A observes a single factor
 86 of their interest x_{e_A} and makes a prediction based on their own belief, c_A : $Y_A =$
 87 $c_A x_{e_A}$ (Fig. 1B). We assume each individual can then independently consult their
 88 AI assistants and revise their prediction based on their reliance on AI, β_A : $Y_A^{re} =$
 89 $\beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Fig. 1B). The revised predictions are then aggregated using two

90 alternative aggregation rules, averaging and clustering, to form collective prediction.
 91 To model AI, we assume that AI has already been trained on similar tasks and
 92 is able to make predictions based on partial or full observations of causal factors
 93 (Fig. 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based
 94 on its internal parameters, which can include factor-specific bias (b_i) and stochastic
 95 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 96 amount of information AI use when making predictions. In one extreme, the AI
 97 can observe all causal factors and use all information to make predictions, regardless
 98 of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 99 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 100 $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 101 and ‘Chatbot AI’, respectively.

102 We assume that appropriate aggregation rule for collective prediction depends
 103 on the type of AI. When Chatbot AI is used, each revised prediction depends on a
 104 single factor. Thus, we use clustering aggregation, under which revised predictions
 105 are first averaged among individuals who are interested in the same factor, and
 106 then summed across factors. When Omniscient AI is used, however, each revised
 107 prediction contains information about all causal factors, in which case clustering
 108 these predictions by interest would be inappropriate. Instead, we use averaging
 109 aggregation, under which revised individual predictions are averaged directly to form
 110 the collective prediction.

111 Each generation consists of multiple round of predictions, and each player is re-
 112 warded a payoff according to the incentive structure (Fig. 1C). For simplicity, we
 113 assume that a sufficiently large number of prediction rounds occurs within each gen-
 114 eration, allowing us to use the expected payoff value rather than computing the
 115 payoffs for every round. We consider two incentive structures from previous work,
 116 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 117 lective accuracy in the absence of AI (Wang et al., 2025). The feedback structure
 118 gives higher payoff to individuals who can push the collective prediction toward the
 119 true target, regardless of their individual predictive accuracy: $\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})$. In
 120 contrast, the niche-expert structure gives higher payoff to individuals who have both
 121 better predictive accuracy and a minority interest: $\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2$. Here, ρ_{e_A}
 122 denotes the proportion of players interested in the same factor, thereby allowing the
 123 niche-expert structure to preserve interest diversity. We note collective intelligence
 124 fails to emerge under the combination of averaging aggregation with niche-expert
 125 structure in the absence of AI (Wang et al., 2025).

126 Through time, the group evolves by allowing each individual to imitate peers
 127 who received higher payoffs (Fig. 1C). Specifically, after each generation, two indi-
 128 viduals are randomly sampled, and one imitates all three strategies (the factor of
 129 interest, factor belief, and reliance on AI) of the other with a probability determined
 130 by the difference in their expected payoffs (Materials and Methods). Individual
 131 predictions are aggregated into a collective prediction $\hat{Y}$ based on the aggregation
 132 rules. Collective accuracy is defined as the reduction in prediction error achieved

by the collective prediction relative to a baseline prediction α_0 (Wang et al., 2025):
 $1 - \mathbb{E}[(Y - \hat{Y}^{\text{re}})^2] / \mathbb{E}[(Y - \alpha_0)^2]$. Hereafter, we say that CI emerges when the collective
accuracy reaches a near perfect level. Full derivations of the collective prediction,
expected payoffs, and collective accuracy under the two AI models are provided in
Supplementary Text S1 and S2.

This model now allows us to explore the impact of AI on CI depending on the
type of AI and the incentive structure. Here, we begin by comparing the emergence
of CI with and without AI and subsequently investigate how different incentive
structures affect the emergence of CI in the presence of AI.

AI can either promote or prevent the emergence of CI

To understand the impact of AI, we first compare collective dynamics with and
without AI. First, consistent with Wang et al. (2025), simple imitation rules and
incentive structures allow collective intelligence to emerge in the absence of AI (Fig.
2A,C,D). One exception is the use of simple averaging with niche-expert incentive
structures, under which the collective accuracy of the population remains near zero.
This is because enhancing individual accuracy causes individuals to predict close to
the true coefficient of their interest, but simply averaging these predictions does not
necessarily produce an accurate estimate of the full prediction target (Fig. S1).

We find that AI use can have heterogeneous outcomes depending on its type.
For example, the use of Omniscient AI can either preserve CI under the feedback
incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
expert incentive structure (Fig. 2B). These results are robust across different levels
of AI accuracy (Fig. S2) and can also be characterized analytically in the large-
population limit (Supplementary Text S3.1).

In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
structure, the use of Chatbot AI leads to a loss of diversity in the population's
interest, making it impossible to account for effect of some causal factors (Fig. S2,
S3). This causes an increase in collective variance and stochastic decrease in collective
accuracy (Fig. 2C; Fig. S3). We note that the loss of interest diversity may arise
even in the absence of AI in finite populations (Fig. S3), but the use of Chatbot
AI makes the group more vulnerable to such loss (Fig. S4). Under the niche-
expert incentive structure, the population becomes overly reliant on AI, causing the
collective prediction to inherit the AI's bias (Fig. S3). This leads to an increase in
collective bias and a decrease in collective accuracy (Fig. 2; Fig. S3). Interestingly,
we find that the niche-expert structure can prevent the use of AI under specific
assumptions about the relationship between the true coefficient and AI bias (Fig.
S2). In this case, the population reduces the reliance on AI assistance, allowing
the CI to emerge eventually. However, this phenomenon can substantially delay the

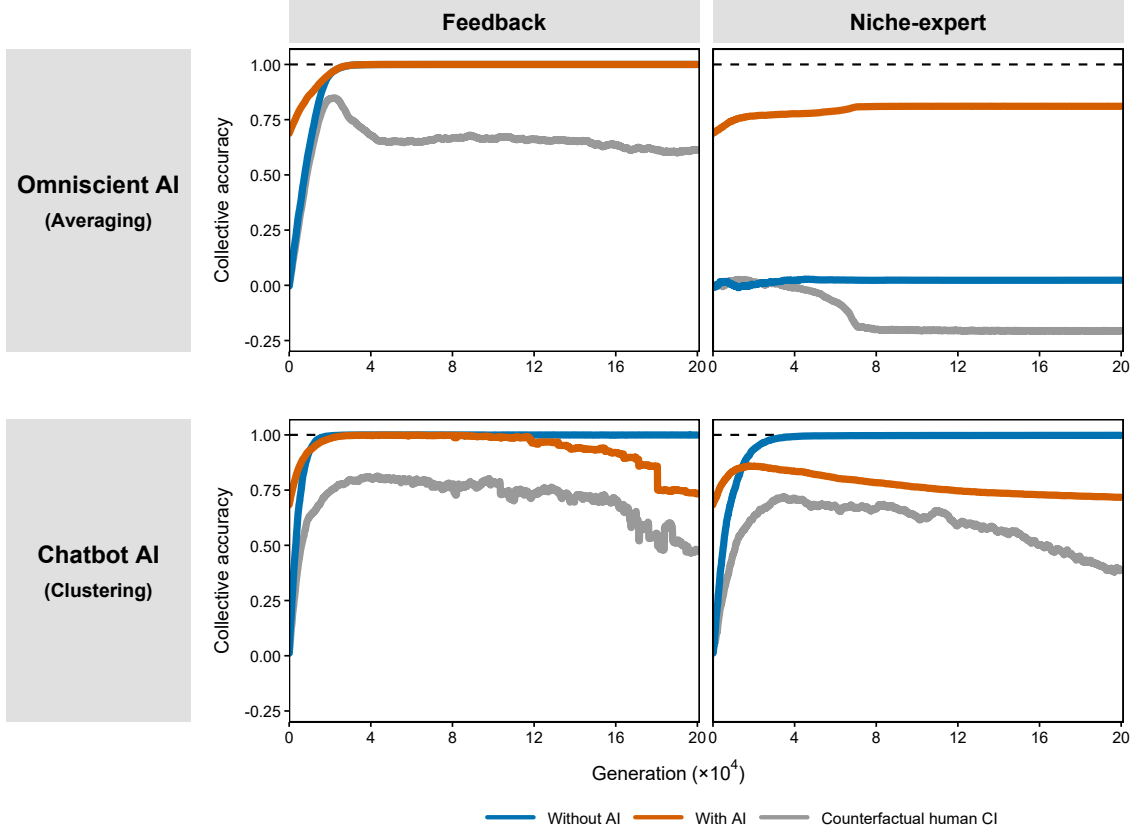


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Blue lines represent trajectories of collective accuracy without AI assistance. Orange lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

173 emergence of CI (Fig. S5).

174 To investigate whether the group can retain its predictive ability when AI assis-
 175 tance suddenly becomes unavailable at any given time, we quantify counterfactual
 176 collective accuracy based on unrevised predictions. Across all scenarios, we find that
 177 the counterfactual collective accuracy is lower than the collective accuracy of the
 178 corresponding population (Fig. 2).

179 In summary, an Omniscient AI can promote CI, but AI operating with limited

information may prevent the emergence of CI. Moreover, reliance on AI can limit inherent ability of the population to make accurate predictions on their own, raising concerns about future AI use. These findings further underscore the need for incentive structures that can mitigate the potential risks associated with AI assistance.

Directly incentivizing or penalizing AI use can additionally harm CI

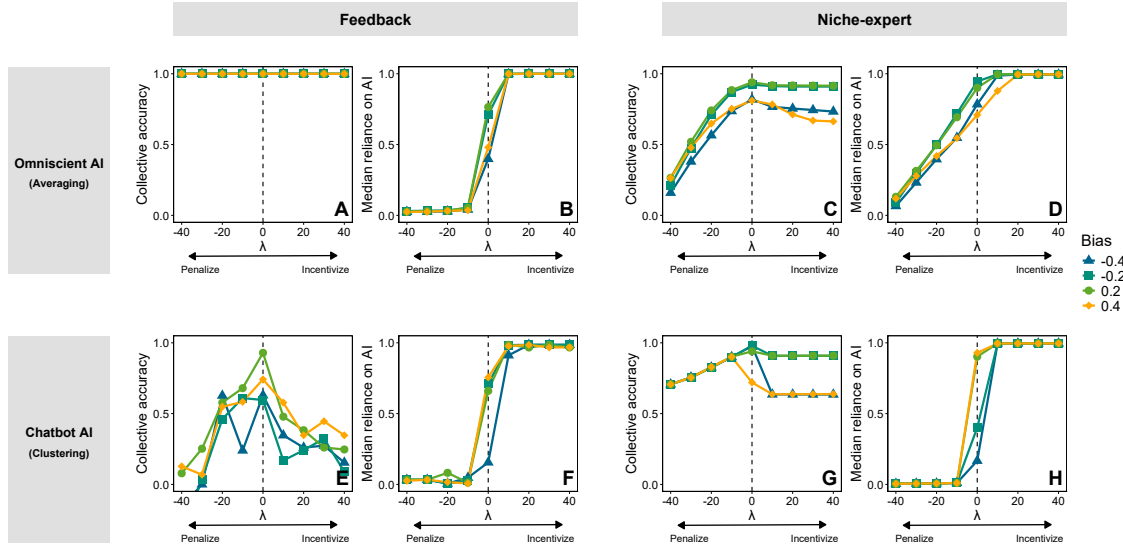


Figure 3: Collective accuracy under the influence of direct intervention on AI use. Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (Omniscient AI, A–D and Chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

Given the strong influence of AI on collective outcomes, a straightforward intervention is to directly incentivize or penalize the use of AI. To investigate whether such direct incentives or penalties can help the emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on

190 AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

191 where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$), and β_A
192 represent individual reliance on AI (Supplementary Text S1.3 and S2.3).

193 Directly incentivizing or penalizing AI use can further reduce the collective ac-
194 curacy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback
195 structure; in the latter case, the population retains collective intelligence regardless
196 of the intervention strength (Fig. 3A). The reduction in collective accuracy arises
197 because incentivizing or penalizing AI use causes individuals to imitate others based
198 on their AI use rather than their predictive accuracy. Thus, stronger interventions
199 can push the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B, D,
200 F, H) but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F,
201 H). When AI bias is relatively large, a weak penalty can lead to higher collective
202 accuracy, but this represents a narrow parameter regime (Figure 3G). Overall, these
203 results indicate that simply incentivizing or penalizing the AI, independent of in-
204 dividuals' performance, use may exacerbate rather than resolve the risk of AI on
205 CI.

206 **Balanced incentives mitigate AI-induced failures of** 207 **CI**

208 Neither the original incentive structures nor simple interventions targeting AI use are
209 sufficient to promote CI. What incentive structure, then, would work in the presence
210 of AI? Here, we propose a “balanced” incentive structure, modeled as the average of
211 the feedback and niche expert incentives (Supplementary Text S2.3).

212 The core idea behind the balanced structure is to maintain interest diversity and
213 prevent collective bias. On one hand, the feedback incentive structure is prone to
214 the loss of interest diversity but can prevent collective bias by incentivizing players
215 whose predictions lie opposite to the collective prediction. On the other hand, the
216 niche-expert structure can maintain interest diversity because it directly incentivizes
217 minority interests. However the niche-expert structure cannot prevent collective bias
218 because, within the same factor of interest, individuals are rewarded only according
219 to their individual predictive performance, which in turn homogenize the predictions.

220 Therefore, when the two incentive structures are combined, the population achieves
221 CI by maintaining interest diversity while limiting collective bias (Fig. 4A, S6). Un-
222 der the balanced structure, individuals remain more evenly distributed across causal
223 factors, and their beliefs lie closer to the true coefficients than under either incentive
224 structure alone (Fig. 4C–E). The balanced structure also produces higher counter-
225 factual collective accuracy than the feedback or niche-expert structure (Fig. 4B,
226 S6). These patterns are robust to how the two incentive structures are weighted
227 when combined (Fig. S7), and are driven by the emergent property of the balanced

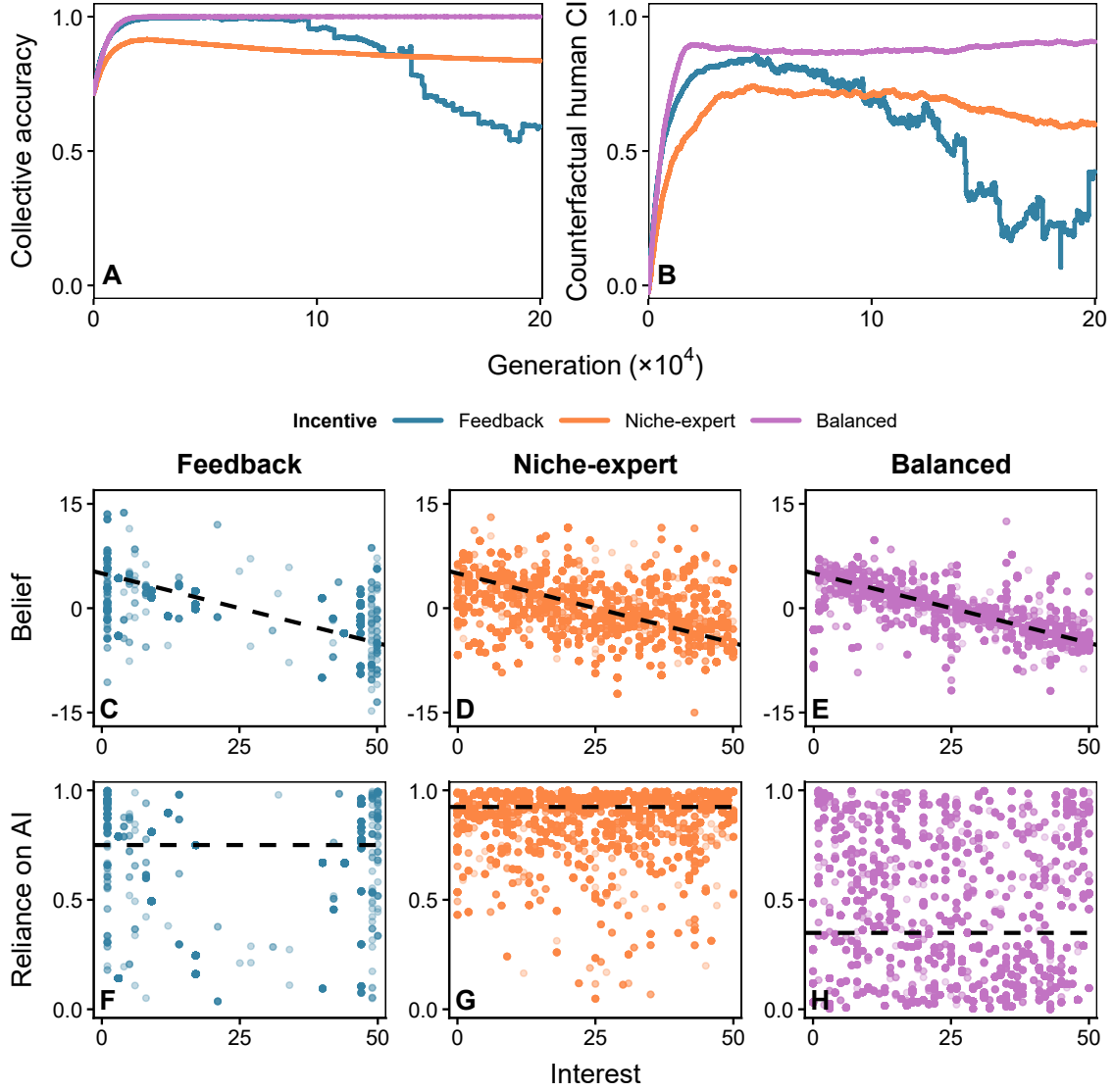


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

228 structure, under which the population increases its reliance on AI as AI becomes
 229 more accurate (Fig S8).

Discussion

The ongoing race towards the development of “better” AI continues to pose questions about the societal benefits and risks of AI use (Bender et al., 2021; Burton et al., 2024; Cui and Yasseri, 2024). Here, we used mathematical models of evolutionary dynamics to investigate how AI use and incentive structures jointly affect the emergence of CI, which is defined as the population’s ability to make accurate predictions. We find that an “Omniscient” AI model, which incorporates information about all relevant causal factors, can improve collective accuracy, whereas a “Chatbot” AI model, whose outcome depends on the user’s interest, can impair it. These contrasting outcomes arise because AI assistance affects not only individual predictions, but also the population’s interest diversity, beliefs, and reliance on AI. We also show that directly incentivizing or penalizing AI use, independent of individual’s performances, can further limit the collective accuracy of the population. Instead, a balanced incentive structure can maintain CI by promoting diversity and limiting bias.

Our analysis reveals that AI can prevent the emergence of CI through two different mechanisms. First, AI use can cause individuals to concentrate on a subset of the factors that determine the prediction target, thereby reducing diversity in their interests. This result is broadly consistent with recent empirical studies showing that AI assistance can improve individual performance but limit the diversity of outputs, in domains ranging from scientific research (Hao et al., 2026) to creative writing (Doshi and Hauser, 2024). Second, over-reliance on AI can cause the population to inherit inherent biases in AI’s predictions (Fügener et al., 2021; Vicente and Matute, 2023), causing the prediction accuracy to decrease eventually. This mechanism can prevent the emergence of CI even when the interest diversity is maintained because biases in AI predictions can be amplified when individual predictions are aggregated. Thus, preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI use need to be calibrated to prevent systematic AI biases from becoming collective biases (Bućinca et al., 2021).

Our analyses of counterfactual collective accuracy also raises an important concern for future AI use: while AI use may improve the apparent performance of individuals and, in some cases, the population as a whole, it can simultaneously reduce the population’s ability to make accurate predictions without AI. These results echo findings from a recent field experiment, which demonstrated that students, who seemingly performed better in solving mathematics problems under AI assistance, performed worse on their own than those who did not have access (Bastani et al., 2025). These findings highlight the need to evaluate AI tools not only by the performance gains they provide during use, but also by whether individuals and populations retain the ability to perform without them (Melumad and Yun, 2025; Ke et al., 2026). Making this distinction clear will be particularly important for education, where AI may help students complete a task without helping them learn how to do it on their own (Bastani et al., 2025; Melumad and Yun, 2025).

As generative AI and large language models are already generating substantial benefits, many organizations, both industrial and academic, are increasingly seeking to improve their collective performance by regulating AI use (Thorp, 2023; Bousquette, 2026; Chawla, 2026; Keary, 2026). Such regulations may take direct forms, including AI-use policies or token-based leaderboards, or may operate indirectly through other incentives, such as productivity rewards or the social stigma associated with AI use (Reif et al., 2025). Our model suggests that directly incentivizing or penalizing AI use is unlikely to induce appropriately calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups to propagate AI biases, whereas penalizing AI use could lead to underreliance, preventing groups from using AI even when it performs well in the relevant domain. Our result instead shows that appropriate AI use and accurate CI can be promoted by focusing on the traditional properties of groups that have been emphasized for decades: maintaining diversity and preventing collective bias, rather than on AI use itself (Hong and Page, 2004; Lorenz et al., 2011; Mann and Helbing, 2017).

There are several limitations to our analyses. First, our model allows for social learning among humans, but AI’s inherent ability remains fixed throughout the evolutionary process. In reality, AI can also learn from human-generated data (Ouyang et al., 2022). Such bidirectional learning increase the similarity between human and AI errors and therefore further reduce the group performance (Steyvers et al., 2022). Second, we model AI errors using fixed, factor-specific additive biases and independent random errors. This assumption allows random errors to be reduced through aggregation, while systematic biases can still persist in the collective prediction. In reality, AI errors may have more complex structures, including biased estimates of factor effects and correlated errors across individuals, factors, or repeated interactions. Further empirical and theoretical work is therefore needed to understand how these error structures affect the emergence of CI. Finally, following established models of CI, we assume that the prediction target is a linear combination of independent causal factors. This assumption does not hold for many real-world tasks, where causal factors may be correlated or interact nonlinearly. Future studies should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

The rapid adoption of AI assistants is changing how individuals learn, work, and make decisions (Tessler et al., 2024; Glickman and Sharot, 2025; Sajadieh et al., 2026; Bick et al., 2026; Ke et al., 2026). While the immediate benefits and risks of these systems have been evaluated at the individual level (Noy and Zhang, 2023; Bastani et al., 2025), our study raises a broader concern that their widespread AI use may reshape collective intelligence of human society. Our results further underline the importance of designing reward structures that preserve minority expertise and prevent AI biases from persisting across the population (Mann and Helbing, 2017; Vicente and Matute, 2023; Glickman and Sharot, 2025; Wang et al., 2025). A detailed empirical understanding of how AI assistance reshapes diversity, expertise, and

reliance at the population level will therefore be critical for predicting the dynamics of future of human collective intelligence.

Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors (Hong et al., 2012; Mann and Helbing, 2017; Wang et al., 2025):

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

where the coefficient α_i determines the magnitude and direction of the effect of factor i , and α_0 represents a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i .

Players

We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel prediction task. In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A . Player A observes the causal factor X_{e_A} and makes a prediction by multiplying the observed factor value by their belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player’s prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that $X_0 = 1$.

When the group is assisted by AI, each player’s strategy additionally includes reliance on AI, denoted by β_A . We assume that each player consults the AI independently and revises their prediction by taking a weighted average of their own prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error.

346 The additive bias associated with factor i is denoted by b_i , and the stochastic error
 347 is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its
 348 biases therefore remain fixed throughout the simulation. For simplicity, we assume
 349 $b_i = b$ for the main analysis for all i . The model formulation nevertheless allows
 350 biases to differ among factors.

351 We assume that the stochastic error follows a Gaussian distribution with mean
 352 zero and standard deviation τ . Each time a player consults the AI, an indepen-
 353 dent error is generated. Errors are therefore independent across players and across
 354 repeated consultations by the same player. Consequently, stochastic AI errors can
 355 be attenuated when independently generated predictions are aggregated across play-
 356 ers. We therefore hold τ constant and vary the magnitude of the additive bias to
 357 manipulate the level of AI accuracy during our analyses.

358 We consider two extreme types of AI that differ in the amount of information
 359 available for prediction. The first is termed the Omniscient AI, which can observe
 360 all M causal factors and incorporate them into its prediction regardless of the user's
 361 interest. The prediction generated by the Omniscient AI for player A is expressed
 362 as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

363 The second is termed the Chatbot AI, which can use only the single causal factor
 364 X_{e_A} associated with the player's interest. The prediction generated by the Chatbot
 365 AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

366 Aggregation rules

367 The revised individual predictions are aggregated to form a collective prediction.
 368 We use two aggregation rules proposed in previous research (Wang et al., 2025),
 369 depending on the type of AI used by the group. First, when the group is assisted by
 370 the Omniscient AI, the AI component of each revised prediction contains information
 371 about all M factors, even though the individual's own prediction is based on only
 372 one factor. Each revised prediction can therefore be treated as an estimate of the
 373 entire target. Accordingly, we calculate the collective prediction by taking the simple
 374 average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (7)$$

375 Second, when the group is assisted by the Chatbot AI, both the player's initial
 376 prediction and the AI's prediction are based on the same causal factor. Each revised
 377 prediction therefore contains information about only one factor and can be treated
 378 as an estimate of that factor's contribution to the target. We consequently use
 379 clustering aggregation.

380 Clustering aggregation is performed by grouping individuals according to their
 381 interests and calculating the mean prediction within each group. The collective
 382 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

383 where $\mathcal{A}_i$ represents the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ represents
 384 the number of players in that set.

385 When the group is not assisted by AI, collective predictions are computed from the
 386 unrevised individual predictions throughout the simulation. Both simple averaging
 387 and clustering can be used in the absence of AI. For fair comparison, we use the
 388 same aggregation rule as in the corresponding AI-assisted condition.

389 Incentive structures

390 Players receive payoffs based on the underlying incentive structure. We consider five
 391 incentive structures: feedback structure, niche-expert structure, direct intervention
 392 on AI use of these two structures, and a balanced structure.

393 The feedback structure does not directly reward individuals according to their
 394 own prediction accuracy. Instead, it rewards predictions that are aligned with the
 395 residual error of the collective prediction and therefore have the potential to move
 396 the collective prediction toward the true value. Given player A 's revised prediction
 397 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

398 The niche-expert structure assigns higher payoffs to individuals who make accu-
 399 rate predictions while having an underrepresented interest. Specifically, the squared
 400 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 401 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

402 Previous research has shown that these two incentive structures can produce near
 403 perfect collective accuracy through social learning, except when the niche-expert
 404 incentive is combined with simple averaging (Wang et al., 2025). They are therefore
 405 used as baseline structures for examining how AI assistance affects the emergence of
 406 CI.

407 To examine the effects of directly encouraging or discouraging AI use, we mod-
 408 ify the feedback and niche-expert structures by adding a term proportional to the
 409 player's reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

Here, λ represents the strength of the direct incentive or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to the original feedback and niche-expert structures.

The balanced structure is defined as the arithmetic mean of the feedback and niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

Simulations

All simulations are conducted using an agent-based model with 50 causal factors ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction target are sampled at the beginning of the simulations and subsequently held fixed.

Coefficients α_i are sampled from a uniform distribution on $[-5, 5]$, except in the simulation presented in Figure 4, in which α_i is set to $5 - 0.2i$ for visualization. Standard deviations σ_i are sampled from a uniform distribution on $[0, 3]$.

Players' initial strategies are also sampled from predefined probability distributions. The distribution of individual belief c_A depends on the aggregation rule. Under simple averaging, each factor's contribution is scaled by the proportion of players interested in that factor. We therefore sample beliefs from a Gaussian distribution with mean zero and standard deviation 100, following the scaling used in previous work (Wang et al., 2025). Under clustering aggregation, beliefs are sampled from a Gaussian distribution with mean zero and standard deviation 5. Because predictions are averaged separately within each interest group and then summed, no additional scaling of c_A is required. The initial interest of each player is sampled uniformly from the integers between 0 and 50, and reliance on AI was sampled from a uniform distribution on $[0, 1]$.

We assume that prediction events occur on a substantially faster timescale than social learning. In each generation, we therefore calculate the expected payoff $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events. The expectation is taken over the distributions of the causal factors $X_1, \dots, X_M$ and the stochastic AI error ϵ , conditional on the current population strategies and the fixed target parameters. Closed form expression of expected payoffs are derived for all incentive structures and are presented in the Supplementary Methods.

Each generation consists of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players, A and B , are randomly sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduce a mutation rate μ , such that with probability μ , player A does

not imitate player B but instead redraw all strategy components from the same distributions used to initialize the population. The mutation rate is set to zero in all simulations except those presented in Supplementary Fig. S4.

Unless otherwise specified, each simulation is run for 200,000 generations. When a result is represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

To evaluate the dynamics of collective prediction, we define six summary metrics: collective accuracy, collective bias, collective variance, counterfactual collective accuracy, interest diversity, and median reliance on AI. At each population state, expectations and variances are evaluated analytically over the distributions of the causal factors and AI errors rather than estimated from sampled prediction events. Collective bias and collective variance constitute the bias–variance decomposition of the group level collective prediction error.

Collective accuracy is normalized relative to the baseline prediction α_0 . A negative value indicates that the collective prediction performs worse than the baseline. Counterfactual collective accuracy is evaluated by setting $\beta_A = 0$ while retaining the evolved interests and beliefs of the AI-assisted population. In calculating interest diversity, we use the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Data and code availability

All data are stored in a publicly available GitHub repository (<https://github.com/parklab-snu/collective-intelligence-with-AI>).

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Contributions

Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology, Software, Visualization, Writing - Original Draft Preparation, Writing - Review & Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervision, Writing - Original Draft Preparation, Writing - Review & Editing.

Competing Interests

The authors declare no competing interests.

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

478 Supplementary Text

479 Here, we derive the expected payoffs and collective accuracy used in the simulations.
480 We consider Omniscient AI and Chatbot AI separately because they contain differ-
481 ent amounts of information and therefore require different aggregation rules. We
482 then analyze the corresponding evolutionary dynamics under a deterministic large-
483 population approximation.

484 S1 Omniscient AI

485 We first consider Omniscient AI, which incorporates all causal factors regardless of
486 the player's interest. We derive individual and collective predictions before calculat-
487 ing the expected payoffs under each incentive structure.

488 S1.1 Model

489 Because Omniscient AI observes all M causal factors, its prediction for player A is
490 given by

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon, \quad (S1)$$

491 where b_i represent factor-specific biases and ϵ represents stochastic error, following
492 a Gaussian distribution with mean zero and standard deviation τ . A new stochastic
493 error is generated independently for each player and prediction event.

494 Let $B = \sum_{i=0}^M b_i$ denote the total additive bias. Because the true prediction
495 target is $Y = \alpha_0 + \sum_{i=1}^M \alpha_i X_i$, the AI prediction can be written as

$$Y^{AI} = Y + B + \epsilon. \quad (S2)$$

496 In contrast to the AI prediction, player A 's initial prediction depends only on the
497 causal factor associated with their interest. Their initial and revised predictions are
498 therefore given by:

$$Y_A = c_A X_{e_A} \quad (S3)$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (S4)$$

499 represents player A 's interest, c_A represents their belief about the effect of the corre-
500 sponding factor, and β_A represents their reliance on AI. As in the main text, we set
501 $X_0 = 1$, such that a player with $e_A = 0$ makes a constant initial prediction.

502 S1.2 Aggregation

503 Because the AI component of each revised prediction contains information about all
 504 M causal factors, we use simple averaging to obtain the collective prediction:

$$\hat{Y}^{re} = \frac{1}{N} \sum_{j=1}^N Y_j^{re} \quad (S5)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j Y^{AI} + (1 - \beta_j) Y_j\} \quad (S6)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j}\} \quad (S7)$$

505 To separate the contributions of AI and the players' initial predictions, we define the
 506 population's mean reliance on AI, the stochastic AI error remaining after aggregation,
 507 and the aggregate contribution of initial predictions associated with each factor:

$$\bar{\beta} = \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S8)$$

$$\bar{\eta} = \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (S9)$$

$$\gamma_i = \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (S10)$$

508 Here, $\bar{\beta}$ represents the mean reliance on AI, $\bar{\eta}$ represents the stochastic AI error
 509 remaining after averaging, and γ_i represents the contribution of players' initial pre-
 510 dictions to factor i . Using these definitions, the revised collective prediction can be
 511 written as follows:

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (S11)$$

512 Thus, the collective prediction consists of an AI-derived component, the contributions
 513 of the players' initial predictions, and the stochastic AI error that remains after
 514 aggregation.

515 S1.3 Expected payoffs

516 We next derive the expected payoff under each incentive structure. Expectations are
 517 taken over the distributions of the causal factors and stochastic AI errors, conditional
 518 on the current population strategies and the fixed parameters of the prediction target.

519 Feedback

520 Under the feedback structure, a player's payoff depends on the alignment between
 521 their revised prediction and the residual error of the collective prediction. To express
 522 this residual, we first define the remaining error in the intercept and in the coefficient
 523 of each causal factor:

$$\delta_0 = \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0), \quad (\text{S12})$$

$$\delta_i = \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0), \quad (\text{S13})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S14})$$

524 where δ_0 represents the remaining error in the intercept, whereas δ_i represents the
 525 remaining error associated with factor i . Because the causal factors and stochastic AI
 526 errors are independent and have zero mean, cross terms vanish when the expectation
 527 is taken. Substituting the revised individual prediction and collective residual into
 528 the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S15})$$

$$= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \right] \quad (\text{S16})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^m \delta_i X_i - \bar{\eta} \right\} \quad (\text{S17})$$

$$= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \quad (\text{S18})$$

$$= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \quad (\text{S19})$$

529 The first term represents the alignment between the AI-derived prediction and the
 530 collective residual, whereas the second term represents the contribution of the player's
 531 initial prediction. The final term arises because the stochastic error in player A 's AI
 532 prediction also contributes to the collective prediction. This finite-population effect
 533 decreases as population size increases.

534 Niche-expert

535 Under the niche-expert structure, the payoff is the negative squared error of the
 536 revised individual prediction, weighted by the frequency of the player's interest.
 537 Because $X_0 = 1$, a player with $e_A = 0$ makes a constant initial prediction. In this

case, the expected payoff is:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S20})$$

$$= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \quad (\text{S21})$$

$$= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \quad (\text{S22})$$

$$= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \quad (\text{S23})$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S24})$$

For a player with $e_A \neq 0$, the initial prediction instead depends on the observed value of factor X_{e_A} . The expected payoff is therefore:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S25})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \quad (\text{S26})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)\alpha_0 + \beta_A B$$

$$+ (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\}^2] \quad (\text{S28})$$

$$= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B\right. \quad (\text{S29})$$

$$+ \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S31})$$

These expressions separate the contributions of the player's belief, the unobserved causal factors, systematic AI bias, and stochastic AI error. Increasing reliance on AI reduces the contribution of errors in the player's initial prediction but increases their exposure to AI bias and stochastic error.

Incentivizing or penalizing AI use

Direct incentives add the term $\lambda\beta_A$ to either the feedback or niche-expert payoff. Because this term is fixed conditional on the player's strategy, its expectation is simply $\lambda\beta_A$. The expected payoffs therefore become

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re})\right] + \lambda\beta_A, \quad (\text{S32})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2\right] + \lambda\beta_A. \quad (\text{S33})$$

Thus, a positive value of λ increases the payoff of strategies with greater reliance on AI, whereas a negative value favors strategies with lower reliance.

551 S1.4 Expected accuracy

552 Using the residual defined above, the mean squared error of the revised collective
553 prediction is:

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S34})$$

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S35})$$

554 The expected collective accuracy is therefore

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S36})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S37})$$

555 The collective prediction error consists of the squared error in the intercept, the re-
556 maining error associated with each causal factor, and the stochastic AI error remain-
557 ing after averaging across players. Counterfactual collective accuracy is obtained
558 from the same expression by setting $\beta_A = 0$ for all players while retaining their
559 evolved interests and beliefs.

560 S2 Chatbot AI

561 Unlike Omniscient AI, Chatbot AI use only the causal factor associated with each
562 player's interest. This causes the information contained in AI predictions to depend
563 directly on the distribution of interests in the population. Here, we derive the collec-
564 tive prediction, expected payoffs, and expected collective accuracy under this form
565 of AI assistance.

566 S2.1 Model

567 The prediction generated by Chatbot AI for player A depends on the causal factor
568 associated with the player's interest:

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon, \quad (\text{S38})$$

569 where b_{e_A} represents factor-specific additive bias, and ϵ represents stochastic error.
570 The stochastic error follows a Gaussian distribution with mean zero and standard
571 deviation τ and is generated independently across players and prediction events. The

572 player's initial and revised predictions are given by Player A's initial and revised
 573 predictions can be then written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S39})$$

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S40})$$

574 Here, c_A represents player A's belief about the effect of their focal factor, e_A repre-
 575 sents their interest, and β_A represents their reliance on AI. Thus, the revised predic-
 576 tion remains specific to the single factor associated with the player's interest.

577 S2.2 Aggregation

578 Because both the player's initial prediction and the Chatbot AI prediction contain
 579 information about the same causal factor, we use clustering aggregation. Revised
 580 predictions are first averaged among players interested in the same factor and then
 581 summed across represented interests:

$$\hat{Y}^{re} = \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \quad (\text{S41})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \} \quad (\text{S42})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon], \quad (\text{S43})$$

582 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number of
 583 players in that set.

584 To simplify the collective prediction, we define the effective coefficient, weighted
 585 AI bias, and stochastic error associated with each interest group:

$$\tilde{c}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \}, \quad (\text{S44})$$

$$\tilde{B}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A}, \quad (\text{S45})$$

$$\bar{B} = \sum_{i=0}^M \tilde{B}_i, \quad (\text{S46})$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon, \quad (\text{S47})$$

$$\bar{\eta} = \sum_{i=0}^M \tilde{\eta}_i, \quad (\text{S48})$$

where $\tilde{c}_i$ represents the mean effective coefficient for factor i after AI assistance, whereas $\tilde{B}_i$ and $\tilde{\eta}_i$ represent its systematic and stochastic error components. Thus, the collective prediction can be written as:

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \tilde{c}_i X_i + \bar{B} + \bar{\eta}, \quad (\text{S49})$$

where $\mathcal{I} = \{i \in \{0, \dots, M\} > 0\}$ denotes the set of represented interests.

S2.3 Expected payoffs

We next derive the expected payoff under each incentive structure. Expectations are taken over the causal factors and stochastic AI errors, conditional on the current population strategies and the fixed parameters of the prediction target.

Feedback

Under the feedback structure, players are rewarded according to whether their revised predictions reduce the error in the collective prediction. We therefore define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - \tilde{c}_0 - \bar{B}, \quad (\text{S50})$$

$$\delta_i = \alpha_i - \tilde{c}_i, \quad (i \neq 0), \quad (\text{S51})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S52})$$

where δ_0 includes the remaining error in the intercept and the total additive AI bias, whereas δ_i measures the remaining error in the coefficient of factor i . Because the causal factors and stochastic errors are independent and have zero mean, substituting the revised individual prediction into the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S53})$$

$$= \mathbb{E}[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \quad (\text{S54})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\}] \quad (\text{S55})$$

$$= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \quad (\text{S56})$$

Thus, the expected feedback payoff depends on the remaining collective error associated with the player's interest, the contribution of factor-specific AI bias to the intercept error, and the player's contribution to the stochastic error of their interest group.

606 Niche-expert

607 Under the niche-expert structure, the negative squared error of each revised indi-
 608 vidual prediction is weighted by the frequency of the player's interest. We consider
 609 $e_A = 0$ separately because these players make constant predictions rather than factor-
 610 dependent predictions. When $e_A = 0$, the expected payoff is

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S57})$$

$$= -\mathbb{E} \left[\rho_0 \left[\left\{ \beta_A \alpha_0 + (1 - \beta_A) c_A \right\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \quad (\text{S58})$$

$$= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0) \beta_A b_0 + \beta_A^2 b_0^2 \right. \quad (\text{S59})$$

$$\left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \quad (\text{S60})$$

611 When $e_A \neq 0$, the player's prediction varies with their focal causal factor. The
 612 expected payoff is then

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S61})$$

$$= -\mathbb{E} [\rho_{e_A} [\left\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \right\} X_{e_A} \quad (\text{S62})$$

$$+ \beta_A b_{e_A} + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i]^2] \quad (\text{S63})$$

$$= -\rho_{e_A} \left\{ (1 - \beta_A)^2 (c_A - \alpha_{e_A})^2 \sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 \right. \quad (\text{S64})$$

$$\left. + \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0 \beta_A b_{e_A} + \beta_A^2 \tau^2 \right\} \quad (\text{S65})$$

613 In both cases, the expected squared error consists of the mismatch between the
 614 player's belief and the true coefficient, contributions from causal factors omitted
 615 from the individual prediction, factor-specific AI bias, and stochastic AI error. The
 616 niche-expert weight ρ_{e_A} scales these errors according to the frequency of the player's
 617 interest.

618 Incentivizing or penalizing AI use

619 Directly incentivizing or penalizing AI use adds the term $\lambda \beta_A$ to the player's payoff.
 620 Because this term is fixed conditional on the player's strategy, it can be taken outside

the expectation. The expected payoffs are therefore

$$\mathbb{E}(\pi_{A,Feedback}) = \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \right] \quad (\text{S66})$$

$$= \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] + \lambda\beta_A, \quad (\text{S67})$$

$$\mathbb{E}(\pi_{A,Niche-expert}) = -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A \right] \quad (\text{S68})$$

$$= -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] + \lambda\beta_A. \quad (\text{S69})$$

Balanced

Under the balanced structure, the payoff corresponds to the arithmetic mean of the feedback and niche-expert payoff components:

$$\mathbb{E}(\pi_A) = \mathbb{E} \left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2} \right] \quad (\text{S70})$$

$$= \frac{\mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] - \mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right]}{2} \quad (\text{S71})$$

The expected payoff under the balanced structure can therefore be calculated directly from the feedback and niche-expert payoffs derived above.

S2.4 Expected accuracy

Finally, we derive the expected collective accuracy under clustering aggregation. The intercept α_0 serves as the baseline prediction. Because the causal factors are independent and have mean zero, its expected squared error is

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S72})$$

Using the expression for the collective residual above, the expected squared error of the revised collective prediction is

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S73})$$

The first two terms represent squared errors in the intercept and factor-specific coefficients. The final term represents the stochastic AI error remaining after predictions are averaged within each interest group. Thus, stochastic error has a larger effect when an interest is represented by fewer individuals. Normalizing this error relative

637 to the baseline gives:

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S74})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S75})$$

638 Here, $R = 1$ represents perfect collective prediction, $R = 0$ represents performance
639 equal to the baseline, and $R < 0$ represents performance worse than the baseline.

640 S3 Evolutionary dynamics

641 Following previous work (Wang et al., 2025; Taylor and Jonker, 1978; Traulsen et al.,
642 2005), we use replicator dynamics as a deterministic large-population approximation
643 to characterize equilibrium states and their local stability. The agent-based sim-
644 ulations themselves follow the Fermi imitation rule defined above. At equal-payoff
645 stationary states, the replicator and Fermi formulations imply the same local invasion
646 directions for any finite selection strength $s > 0$, although their transient dynamics
647 need not coincide. We focus on the feedback and niche-expert structures, for which
648 the long-term dynamics can be characterized analytically. The direct interventions
649 on AI use and the balanced structure are examined through simulations.

650 Let λ denote a strategy, p_λ its frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff.
651 Then, the corresponding replicator equation can be written as:

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S76})$$

652 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$ represents the mean expected payoff of the population. At an
653 internal stationary state, all strategies with positive frequency must have the same
654 expected payoff. A boundary state can also be stationary because a strategy with zero
655 frequency cannot increase unless it is introduced into the population. We therefore
656 assess the stability of boundary states by asking whether they can be invaded by a
657 rare alternative strategy.

658 S3.1 Omniscient AI

659 With Omniscient AI, every AI prediction contains information about the full predic-
660 tion target, and revised predictions are combined through averaging. As population
661 size increases, independent AI errors vanish from the collective prediction. How-
662 ever, these errors remain in individual prediction errors and can therefore continue
663 to affect niche-expert payoffs. We consider the two incentive structures separately.

664 S3.1.1 Feedback

665 Under the feedback structure, both the stochastic error in the collective prediction,
 666 $\bar{\eta}$, and its contribution to expected payoff vanish as $N \rightarrow \infty$. Following previous
 667 work (Wang et al., 2025), we define the Lyapunov function as the expected error of
 668 collective prediction:

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S77})$$

669 The function V is non-negative and equals zero when the collective prediction is
 670 identical to the target. To determine how V changes over time, we take its time
 671 derivative:

$$\frac{dV}{dt} = -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \quad (\text{S78})$$

$$= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \quad (\text{S79})$$

$$= -2 \left[\left\{ (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right]. \quad (\text{S80})$$

672 For notational convenience, we define

$$K = (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S81})$$

673 Although K changes with the population state, it is common to all strategies at a
 674 given state and can therefore be treated as a constant with respect to the strategy
 675 index λ . We also define:

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S82})$$

676 Under the replicator equation, the rate of change in the population mean of a
 677 strategy component is given by its covariance with expected payoff.

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S83})$$

678 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta)cG_e, u). \quad (\text{S84})$$

679 Therefore,

$$\frac{dV}{dt} = -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta)cG_e, u) \} \quad (\text{S85})$$

$$= -2 \text{Cov}(K\beta + (1 - \beta)cG_e, u). \quad (\text{S86})$$

For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy λ is

$$u_\lambda = K\beta_\lambda + (1 - \beta_\lambda)c_\lambda G_{e_\lambda}. \quad (\text{S87})$$

This expected payoff is identical to the quantity inside the covariance above. Therefore,

$$\frac{dV}{dt} = -2\text{Cov}(u, u) = -2\text{Var}(u) \leq 0. \quad (\text{S88})$$

Because payoff variance is non-negative, the expected collective prediction error cannot increase and decreases whenever the strategies currently represented in the population differ in their expected payoffs. The derivative becomes zero when all represented strategies receive the same expected payoff. However, this condition does not necessarily imply that $V = 0$, because a population can lack strategies that would improve its collective prediction.

If $V > 0$, at least one component of the collective residual remains nonzero. A rare strategy whose prediction is aligned with this residual can obtain a higher expected payoff than the resident strategies and invade the population. Thus, provided that novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$. At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition arises when all represented strategies receive the same expected payoff. However, this condition does not necessarily imply that $V = 0$, because a population can lack strategies that would improve its collective prediction.

If $V > 0$, at least one component of the collective residual remains nonzero. A rare strategy whose prediction is aligned with this residual can obtain a higher expected payoff than the resident strategies and invade the population. Thus, provided that novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$. At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

S3.1.2 Niche-expert

Unlike feedback payoffs, niche-expert payoffs depend on the prediction error of each individual. Stochastic AI error therefore remains in the expected payoff even when $N \rightarrow \infty$, and we retain $\tau > 0$ throughout this analysis.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S89})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its

unconstrained stationary point is then:

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S90})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

To compare the optimized strategies across interests, let L_e denote the minimum expected individual squared error within interest e , $\mathbb{E}[(Y_A^{re} - Y)^2]$. For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S91})$$

If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating the expected squared prediction error at the corresponding boundary.

Interest 0 must be considered separately because $X_0 = 1$ and the player's belief determines a constant prediction. Optimizing with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S92})$$

Substituting this into the expected payoff gives a strictly concave function of β , with the unique optimum:

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S93})$$

Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S94})$$

and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S95})$$

At this optimum, the player adjusts their belief to offset the systematic AI bias. The stochastic component of AI error cannot be removed in the same way and therefore limits the benefit of AI reliance.

We denote the optimal strategy within interest e by S_e . When $\beta_e^* = 1$, different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class. A stable state can contain only the optimal strategy class within each represented interest because a suboptimal strategy can be invaded by S_e .

739 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
740 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S96})$$

741 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator equa-
742 tion at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S97})$$

743 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S98})$$

744 This internal stationary state is locally stable. At equilibrium, all represented in-
745 terests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
746 interest increases slightly above its equilibrium value, its payoff decreases because
747 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of an
748 interest decreases below its equilibrium value, its payoff increases. The replicator dy-
749 namics therefore generate a restoring force that reduces small deviations in interest
750 frequencies and drives the population back toward the internal stationary state.

751 If one or more interests are absent, the resulting boundary state can be stationary
752 under the replicator equation, but it is unstable against invasion. The payoff of a rare
753 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
754 population is negative when $L_e > 0$, so the rare interest increases when introduced
755 at sufficiently small frequency.

756 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
757 prediction strategies and their frequencies. Whether the constrained optimum has
758 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
759 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
760 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
761 interest frequency remain uniquely determined.

762 Having characterized this equilibrium, we next compare its collective prediction
763 with a prediction made entirely by AI. In particular, the expected squared error of
764 an Omniscient AI prediction is

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S99})$$

765 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
766 fore, because L_e is the minimum individual expected squared prediction error within
767 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S100})$$

Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S101})$$

Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S102})$$

The collective prediction is the average of individual revised predictions. By convexity of the squared error,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S103})$$

Therefore,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] < \mathbb{E} \left[(Y - Y^{\text{AI}})^2 \right]. \quad (\text{S104})$$

Thus, the diversity of optimized predictions allows the population to outperform an individual AI prediction at equilibrium.

However, outperforming AI does not imply perfect collective accuracy. Under averaging, the non-AI contribution associated with a particular factor is provided only by individuals interested in that factor. The predictions of individuals with other interests therefore attenuate that factor's coefficient in the collective prediction. For a nonzero factor i , the residual coefficient of the stationary collective prediction can be written as

$$\delta_i^* = \alpha_i \left\{ 1 - \bar{\beta}^* - \rho_i^* (1 - \beta_i^*) \right\} = \alpha_i \sum_{e \neq i} \rho_e^* (1 - \beta_e^*). \quad (\text{S105})$$

At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S106})$$

Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error. Hence,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] > 0, \quad (\text{S107})$$

and consequently the stationary collective accuracy satisfies $R^* < 1$. Thus, the niche-expert structure preserves interest diversity and allows the population to outperform AI, but averaging prevents the population from reaching perfect collective accuracy.

Supplementary Figure

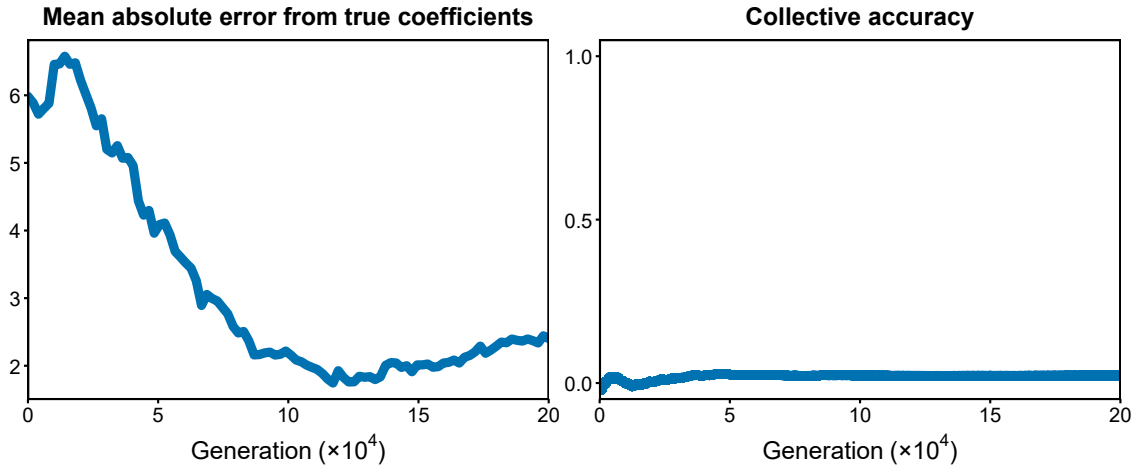


Figure S1: **Niche-expert structure with averaging aggregation cannot promote CI without AI.** Trajectories of the mean absolute error from true coefficients and collective accuracy under the niche-expert structure with averaging aggregation in the absence of AI. The mean absolute error from true coefficients is calculated by first taking, for each interest, the absolute difference between the true coefficient and the mean belief of individuals sharing that interest, and then averaging these differences across all $m + 1$ interests. The mean absolute error decreases across generations, indicating that the mean beliefs converge toward their corresponding true coefficients. However, collective accuracy remains near zero, indicating that simply averaging individual predictions does not yield accurate collective predictions. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$.

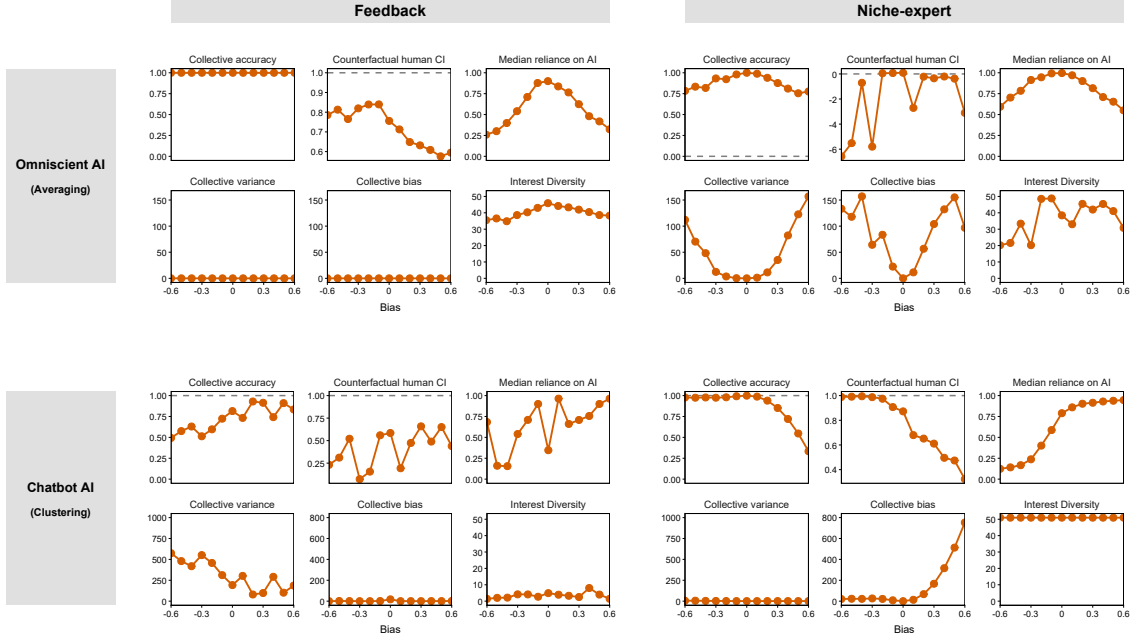


Figure S2: **Evaluation metrics across various levels of AI bias.** Evaluation metrics across varying levels of AI bias under different assumptions about AI types (Omniscient AI, and Chatbot AI) and incentive structures (feedback, niche-expert). Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, and counterfactual human CI represents the group’s predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as Hill-Shannon diversity of interest within the population, median reliance on AI represents the median value of reliance on AI (β). Collective bias represents systematic deviation of the collective prediction from the true target, collective variance represents variability of the collective prediction error across prediction events. All the evaluation metrics are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$.

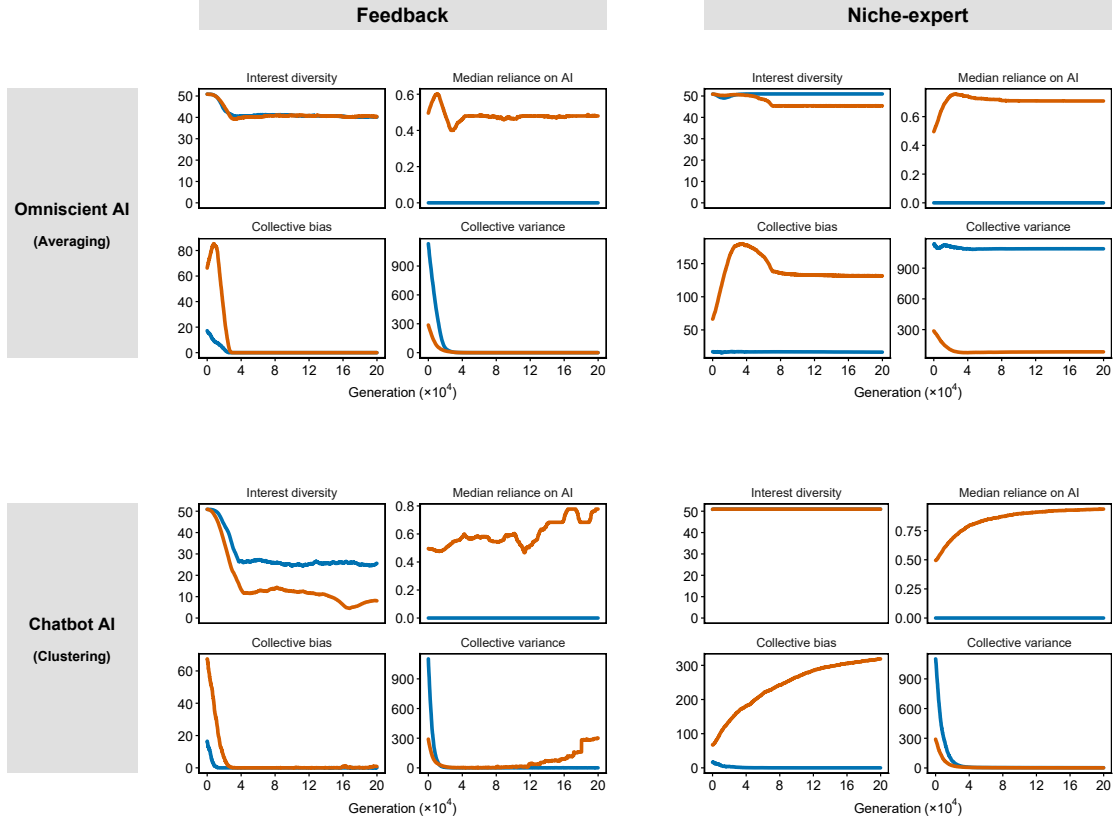


Figure S3: **Comparison of evaluation metrics with and without AI.** Trajectories of evaluation metrics when the group is assisted by the Omniscient AI and Chatbot AI. Here, feedback and niche-expert incentive structures are considered. Blue lines represent trajectories of evaluation metrics without AI assistance. Orange lines represent trajectories of evaluation metrics with AI assistance. Interest diversity is define as hill-shannon diversity of interest within population, median reliance on AI represent population median of reliance on AI (β), collective bias represent systematic deviation of the collective prediction from the true target, collective variance repressent variability of the collective prediction error across prediction events. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

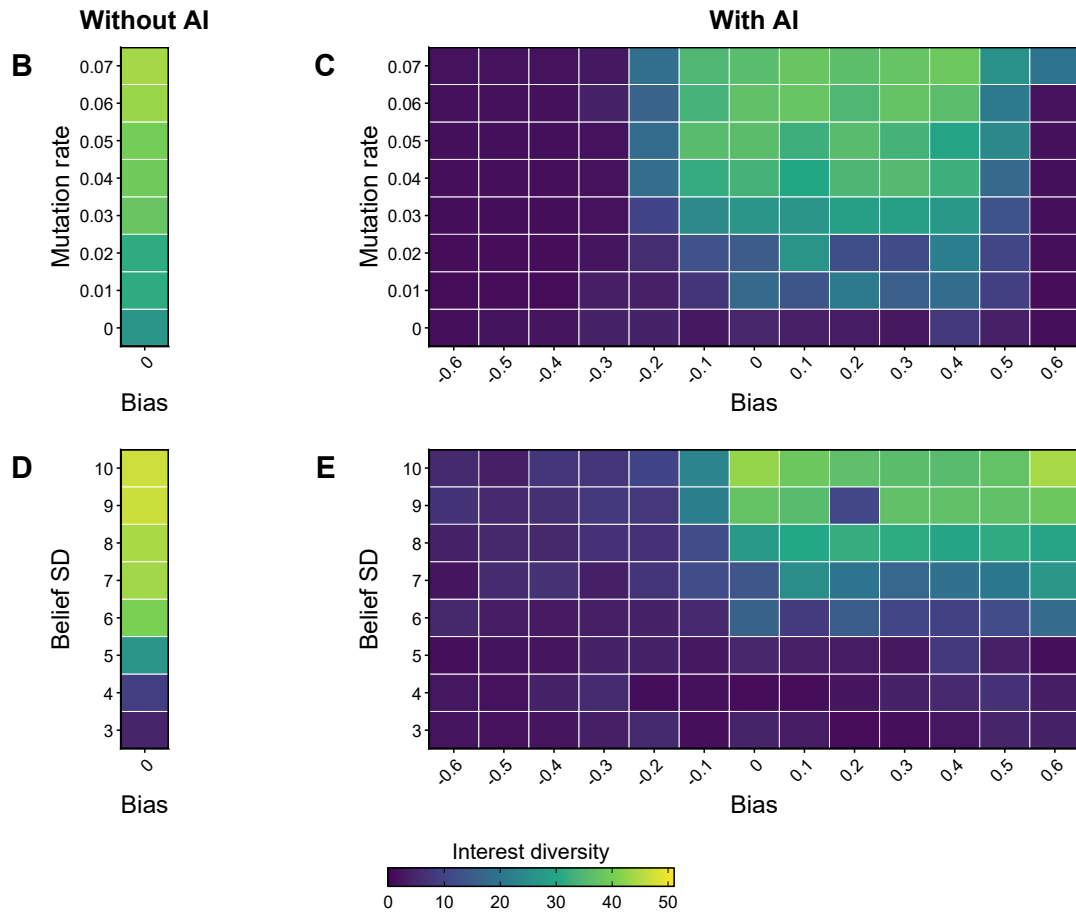
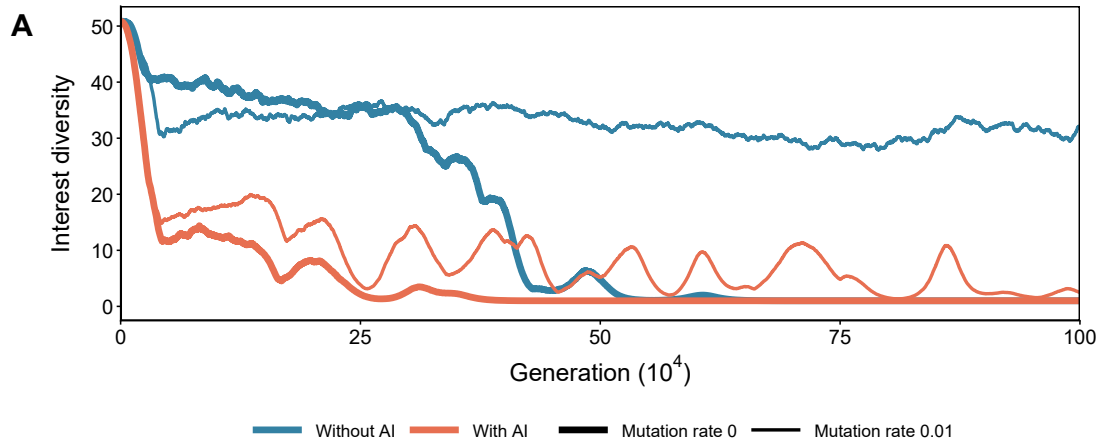


Figure S4: **Feedback structure becomes more prone to interest diversity loss with AI** Trajectories of interest diversity with and without AI under the feedback structure (A). Heatmaps of interest diversity across varying levels of mutation rate (B, C) and the standard deviation of the initial belief distribution (D, E), with and without AI. Interest diversity declines toward zero even without AI over long timescales when there is no mutation and the initial belief distribution is narrow. However, with AI, interest diversity declines more strongly, even under moderate mutation rates and wider initial belief distributions. Interest diversity in the heatmaps is averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$ for (D, E), $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$ for (A, B, C), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$, Chatbot AI is used.

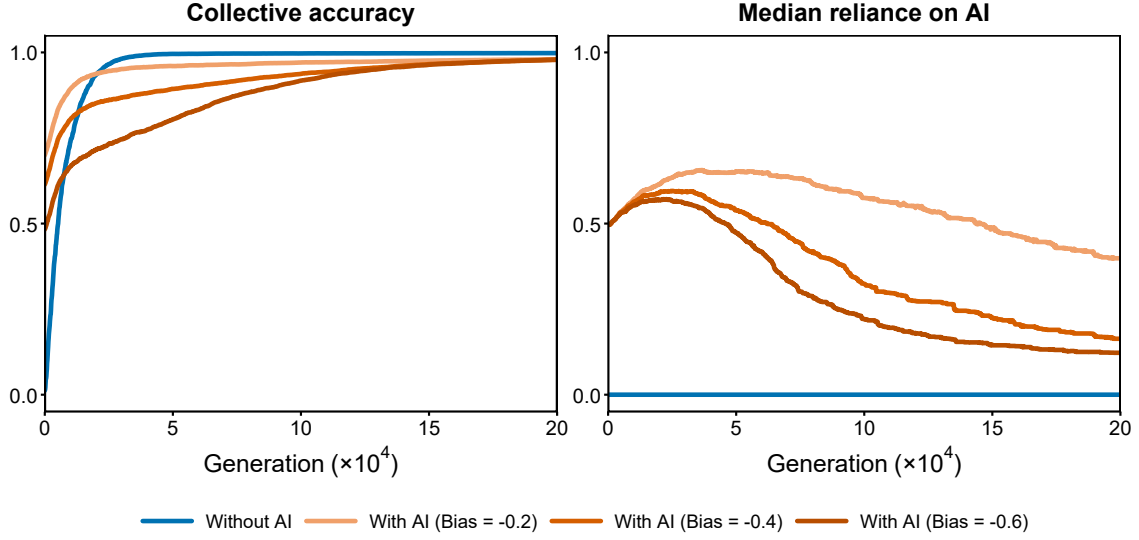


Figure S5: **Niche-expert structure may reduce reliance on AI, delaying the emergence of CI.** Trajectories of collective accuracy and median reliance on AI when the niche-expert structure causes reliance on AI to decline over time. Red lines represent trajectories with AI across varying levels of AI bias (-0.2, -0.4, and -0.6). Green lines represent trajectories without AI; reliance on AI is set to zero in the absence of AI. This reduction in reliance on AI occurs when the α_0 and AI bias have opposite signs, and is associated with the eventual emergence of CI, although CI emerges substantially later than in the condition without AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

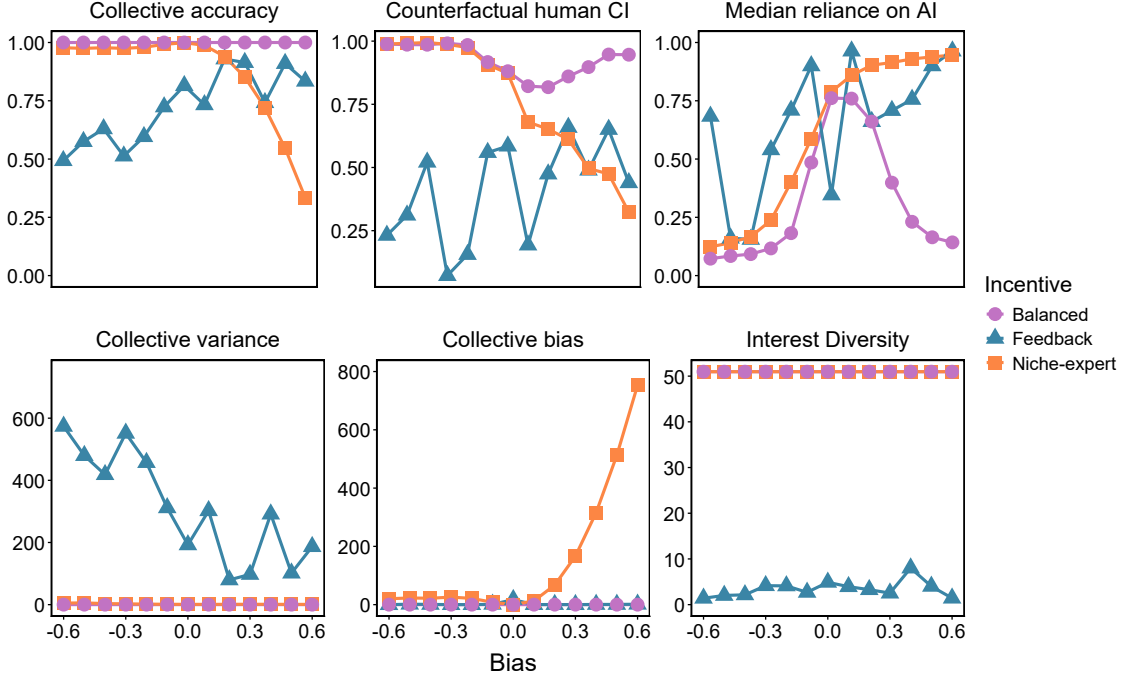


Figure S6: **Comparison of evaluation metrics between incentive structures across various levels of AI bias.** Evaluation metrics under feedback, niche-expert and balanced structures across varying levels of AI bias. Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. All the evaluation metrics are averaged over generations 190,000–200,000. The balanced structure maintains near-perfect collective accuracy across bias levels and higher counterfactual human CI than the other incentive structures. These patterns are driven by well-calibrated reliance on AI, which increases as AI becomes more accurate, and by well-preserved interest diversity, which together lead to low collective variance and collective bias. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

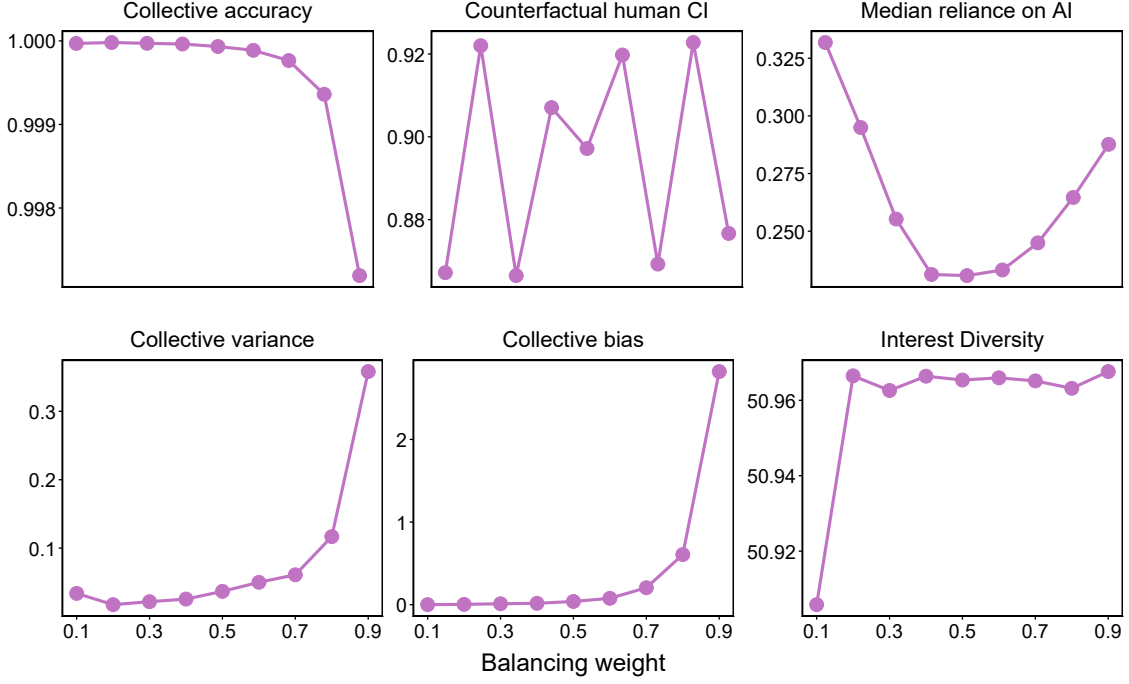


Figure S7: **Balanced structure is robust across varying balancing weights.** Evaluation metrics across varying levels of the balancing weight (0.1 – 0.9). The balancing weight represents the weight assigned to the niche-expert structure when constructing the balanced structure by combining the niche-expert and feedback structures. A balancing weight of 0.5 represents equal weighting of the niche-expert and feedback structures, whereas 0 corresponds to the feedback structure and 1 corresponds to the niche-expert structure. Although the evaluation metrics vary as the balancing weight changes from 0.1 to 0.9, the absolute magnitude of these deviations remains limited. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$, $\tau = 0.3$, Chatbot AI is used.

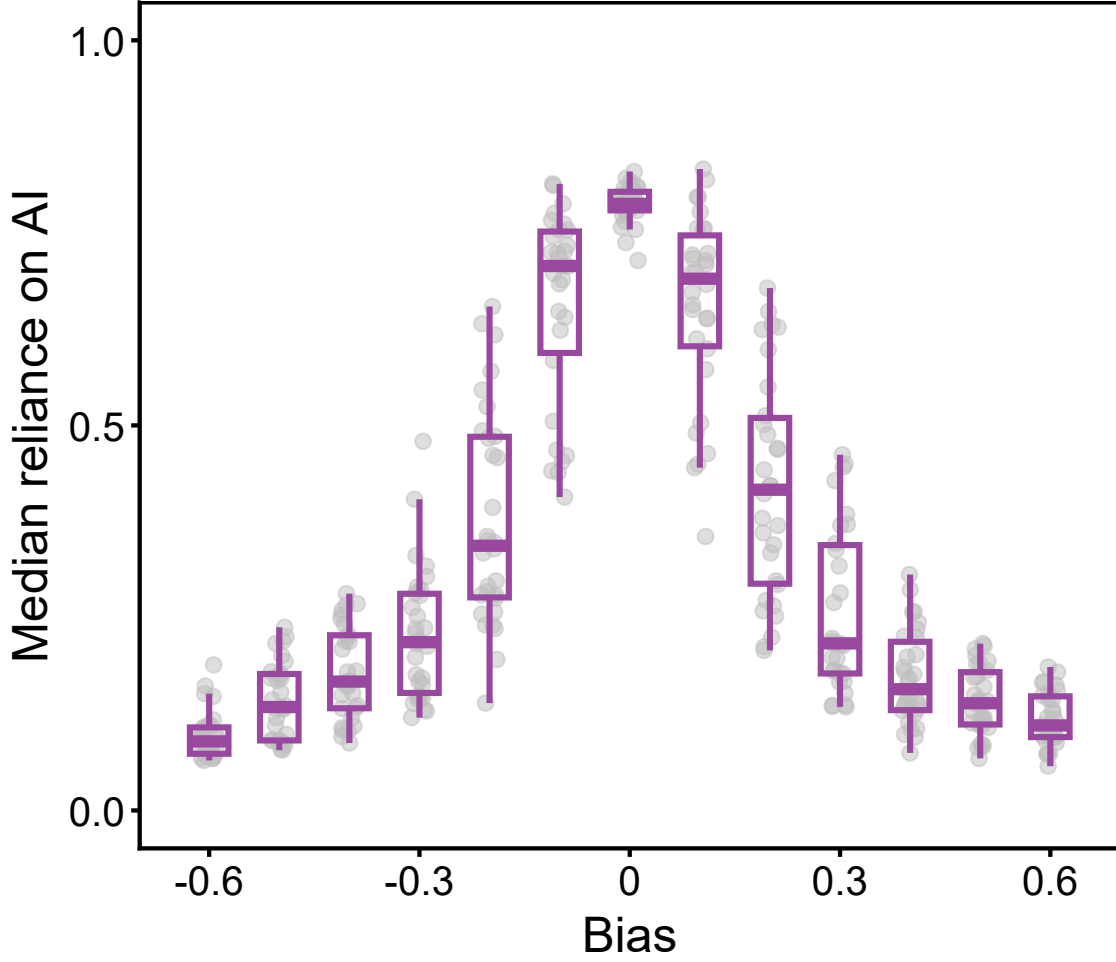


Figure S8: **Balanced structure shows well-calibrated reliance on AI.** Median reliance on AI across varying levels of AI bias. Thirty randomly initialized simulations were performed for each bias level. Each dot represents a single simulation, and the box plots show the median and interquartile range (IQR) for each bias level. The balanced structure shows well-calibrated reliance on AI, with reliance increasing as AI bias decreases. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

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